

PERFORMANCE OF KALINA CYCLE SYSTEM 11; (KCS11) USING LOW TEMPERATURE HEAT SOURCES

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ABSTRACT

With the ever increasing demand for energy, exploiting low-temperature heat sources has seen significant interest recently. The conventional Organic Rankine Cycle (ORC) is a typical approach used to exploit low temperature heat sources, but suffers from low efficiency. The Kalina cycle is a reversed absorption cooling system that uses an ammonia-water binary mixture as the working fluid. This paper investigates the performance of Kalina Cycle System 11 (KCS11), which utilizes low temperature heat sources below 200°C compared to ORC based on pure ammonia and R134a. Thermal modeling of KCS11 and ORC cycles was carried out using conservation of energy and mass laws applied to each component and the resulting set of equations were solved using the Engineering Equation Solver (EES) software. The cycles performance was investigated at various operating conditions including evaporator pressure of 15bar-32bar, heat source temperatures of 333K to 473K, heat sink temperature of 283K and in the case of KCS11 various ammonia mass fractions at the evaporator outlet. Results show that the Kalina cycle KCS11 can produce up to 40% increase in the efficiency compared to ammonia and up to 20% increase in compared to R134a.

1. INTRODUCTION

With the increasing demand for and cost of energy, the exploitation of low-grade heat sources such as geothermal, solar and waste heat sources has been receiving more attention. Through the advancement of technology, there is much interest in designing more efficient, reliable and cost-effective energy conversion systems that will provide means of exploiting low temperature heat sources that might not otherwise be utilized. The Kalina cycle and ORC provide a couple of possible solutions to the problem of recovering the low temperature energy that is usually thrown away in industrial waste heat but ORC has a drawback in terms of the overall efficiency (Madhawa et al, 2007). Interest in the Kalina cycle has been growing since it was patented by Dr. Alexander Kalina in the 1980s. The Kalina cycle is a modified conventional ORC or reversed absorption cycle (Mlcak H A, 1996). The Kalina cycle is the first major advance in power generation technology over the Rankine cycle that was invented by William Rankine of Scotland over 150 years ago. Compared to conventional thermodynamic cycles, the Kalina cycle power plant can offer an efficiency gain of 10 - 50% for low temperature thermal energy sources such as geothermal brine at 60°C to 200°C (Lolos and Rogdakis, 2009), waste heat from gas turbines (Jonsson and Yan, 2000, Senthil Murugan and Subbarao, 2008) and waste heat from iron and steel industry (Johnson, 2003). It is likely that Kalina cycle plants can cost even less to build than Rankine cycle plants with an equal output. Up to 30% savings for applications with low temperature heat sources and up to 10% savings for direct fired or bottoming cycle plants have been reported by Global Geothermal Limited (Lolos and Rogdakis, 2009).

Generally, there are different types of Kalina cycle families, which are known by their unique names. For example, KCS5 is particularly applicable to direct fired plants. KCS6 is applicable to gas turbine based on combined cycles and KCS11 and KCS34 are designed for exploiting low temperature heat sources. For this work KCS11 was selected as it is the most applicable for low grade waste heat sources at temperatures between 100°C and 200°C (Galanis et al, 2009). These temperatures can be obtained from commercially available solar thermal collectors such as evacuated tubes and compound parabolic collectors. In this paper a theoretical analysis of the KCS11 cycle using ammonia/water working pair and the ORC cycle based on pure ammonia or pure R134a to investigate the performance of the two cycles at various operating conditions including heat source temperature ranging from 333K to 463K, heat sink temperature of 283K, evaporator pressure ranging from 10bars to 32 bars and ammonia concentration at the exit of the evaporator ranging from 0 to 1. R134a was recommended by Tchanche B F et al 2009, to

be one of the most suitable working fluids for small scale solar powered ORC systems while NH_3 is a natural refrigerant with no ozone depletion potential or global warming effect.

2. KCS11 AND ORC THERMAL MODELLING

Figure 1 (a) shows a schematic diagram of the Kalina cycle KCS11. It consists of a turbine, absorber, condenser, evaporator, separator, regenerator, pump and throttling valve. In the evaporator, the ammonia / water mixture is heated up by the low temperature heat source and then passes to the separator. In the separator, the saturated vapour part of the mixture separates from the liquid. The rich ammonia saturated vapour mixture then expands through the turbine producing power output and then it passes through the absorber. The ammonia/water solution leaves the absorber to the condenser where it is condensed and then it is pumped to increase its pressure to that corresponding to the evaporator. The hot weak ammonia/water saturated liquid mixture leaving the separator is then directed to the regenerator where it is cooled by the rich ammonia mixture flowing back to the evaporator. After the regenerator, the weak ammonia solution passes through a throttling valve to lower its pressure. The organic Rankine cycle consists of four components namely; the turbine, evaporator, condenser and pump as shown in figure 1(b). In the ORC modelled in this paper, pure ammonia and R134a were used as the working fluid.

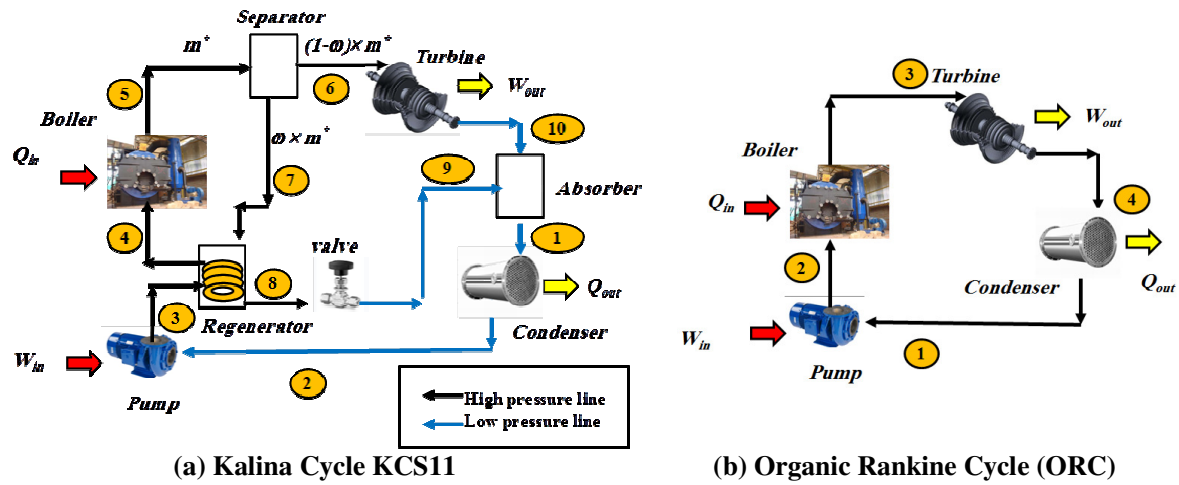


Figure 1: Flow diagram of different cycles

The modelling of the KCS11 is carried out by applying the steady flow energy and mass balance equations to the various components of the system neglecting changes to kinetic and potential energy and frictional losses. Assuming that both the pump (η_{pump}) and turbine (η_{turb}) have isentropic efficiency of 80%, the specific work required by the pump (w_{pump}) and the specific work produced by the turbine (w_{turb}) were calculated by (Eqs 1-2).

$$w_{\text{pump}} = (v_2 * (P_3 - P_2)) / \eta_{\text{pump}} = (h_3 - h_2) \quad \text{J/kg} \quad [1]$$

$$w_{\text{turb}} = \left((1 - \omega) * \frac{(h_6 - h_{10,s})}{\eta_{\text{turb}}} \right) = (1 - \omega) * (h_6 - h_{10}) \quad \text{J/kg} \quad [2]$$

Where ω is the ratio of mass flow rate of the ammonia weak solution leaving the separator to the regenerator (state 7) and the mass flow rate of the ammonia rich solution entering the separator (state 5). v_2 , h_6 and h_{10} are the specific volume at pump inlet, specific enthalpy at turbine inlet and specific enthalpy at turbine outlet obtained as function of temperature, pressure and ammonia concentration in the solution. $h_{10,s}$ is the specific enthalpy of the ammonia / water solution assuming isentropic expansion through the turbine. In all the modelling, the dryness fraction at the exit of the turbine was limited to 90% to minimise liquid droplet formation in the turbine.

The pressure reducing valve after the regenerator is assumed to be adiabatic, thus the enthalpy of the fluid at inlet is equal to that at exit of the valve.

$$h_9 = h_8 \quad \text{J/kg} \quad [3]$$

The separator and the absorber are assumed to be adiabatic with no external heating or cooling applied.

$$h_5 - h_6 = \omega * (h_7 - h_6) \quad \text{J/kg} \quad [4]$$

$$h_1 - h_{10} = \omega * (h_9 - h_{10}) \quad \text{J/kg} \quad [5]$$

For the regenerator, assuming no heat losses to the surrounding and a minimum temperature difference (Pinch point) of 4K, the rate of energy absorbed by the ammonia rich solution (state 3 to state 4) is equal to the heat lost by the ammonia weak solution (state 7 to state 8) thus:

$$h_4 - h_3 = \omega * (h_7 - h_8) \quad \text{J/kg} \quad [6]$$

For the evaporator and condenser, the specific energy absorbed from the heat source and that rejected to the heat sink are given by:

$$q_e = h_5 - h_4 \quad \text{J/kg} \quad [7]$$

$$q_c = h_1 - h_2 \quad \text{J/kg} \quad [8]$$

The thermal efficiency of the KCS11 cycle can then be determined from:

$$\eta = (w_{\text{net}}/q_e) * 100 \quad \% \quad [9]$$

Where the net power output is determined by:

$$w_{\text{net}} = (1 - \omega)(h_6 - h_{10}) - (h_3 - h_2) \quad \text{J/kg} \quad [10]$$

The modelling of ORC has been conducted in order to compare the KCS11 performance to those of the ORC. Applying the steady flow energy equation on the components of the ORC cycle, the net power output, the evaporator heat input and the cycle thermal efficiency can be determined using Eqs 11-15.

$$w_{\text{pump}} = (v * (P_2 - P_1))/\eta_{\text{pump}} = (h_2 - h_1) \quad \text{J/kg} \quad [11]$$

$$w_{\text{turb}} = \frac{(h_3 - h_{4,s})}{\eta_{\text{turb}}} = h_3 - h_4 \quad \text{J/kg} \quad [12]$$

$$w_{\text{net}} = (h_3 - h_4) - (h_2 - h_1) \quad \text{J/kg} \quad [13]$$

$$q_e = h_3 - h_2 \quad \text{J/kg} \quad [14]$$

$$\eta = (w_{\text{net}}/q_e) * 100 \quad \% \quad [15]$$

The modelling was conducted using Engineering Equation Solvers (EES) where pure ammonia and pure R134a thermodynamic properties are available. Also, the ammonia / water mixture properties are based on a formulation by Ibrahim and Klein, 1993 in the external library of EES software. In all the modelling performed, the minimum temperature difference between the working fluid and the heat source in the evaporator or heat sink in the condenser was assumed to be 4K.

3. RESULTS AND DISCUSSIONS

This work investigated the performance of KCS11 compared to ORC in terms of its effectiveness to utilize commercially available solar thermal collectors that produce heat at temperatures of 100-200°C. The ORC utilized either pure ammonia or pure R134a as a working fluid while the KCS11utilised ammonia / water mixture.

Figures 2, 3, 4 and 5 show the thermal efficiency curves for the KCS11 as a function of the ammonia mass fraction at the evaporator outlet for several heat source temperatures. In the graphs, the heat sink temperature was set at 283K and the evaporator pressure varied from 10 bars (figure 2), 15 bars (figure 3), 23.5 bars (figure 4) to 32 bars (figure 5). The results show that as the heat source temperature increases, the maximum thermal efficiency of the Kalina Cycle increases. Also, the results show that when the ammonia concentration is too lean in the working fluid; the thermal efficiency of the cycle drops quickly. This trend can be explained as follows. At specific temperature and pressure, as the concentration of ammonia decreases, the mixture becomes saturated or even subcooled liquid. Thus there will be no vapour generated during the separation process hence the turbine work output becomes negligible and the efficiency reduces steeply. On the other hand as the ammonia mass fraction increases, the thermal efficiency of the cycle drops gradually. This indicates that; for a KCS11 in operation the

ammonia mass fraction of the working fluid need to be rich to avoid an entire loss in the thermal efficiency of the cycle. Thus; to keep the cycle at a reasonable efficiency and stable operating condition; the mass fraction of ammonia should be at the range of 0.55 to 0.9. The graphs also show that, the point of maximum efficiency shifts towards low concentration values by increasing the evaporator temperature (heat source).

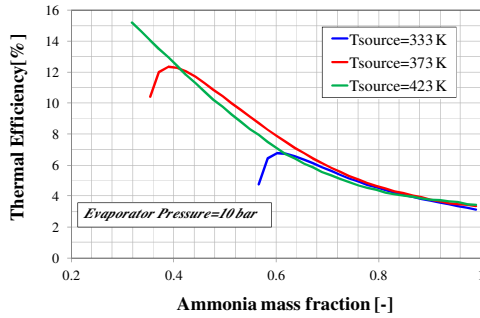


Figure 2: Thermal efficiency of KCS11 with a evaporator pressure of 10 bars and a sink temperature of 283K

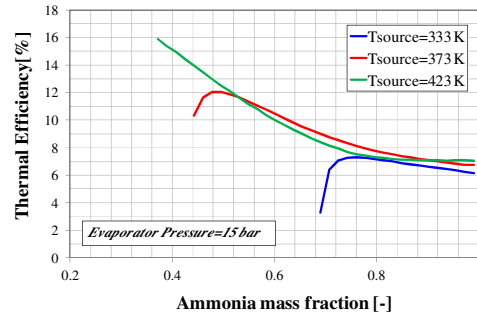


Figure 3: Thermal efficiency of KCS11 with a evaporator pressure of 15 bars and a sink temperature of 283K

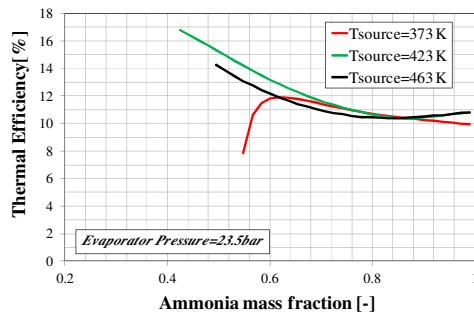


Figure 4: Thermal efficiency of KCS11 with a evaporator pressure of 23.5 bars and a sink temperature of 283K

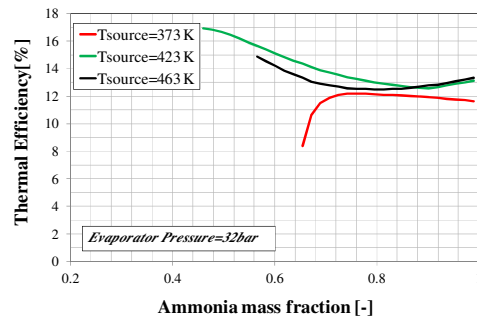


Figure 5: Thermal efficiency of KCS11 with a evaporator pressure of 32 bars and a sink temperature of 283K

Figure 6(a) and 6(b) shows the thermal efficiency of ORC using pure ammonia and pure R134a as the working fluid. To compute the thermal efficiency of the cycle, the evaporator pressure was increased while holding the heat sink and heat source temperatures constant. The evaporator pressure was limited so that the turbine exit quality was no less than 90%. It is clear from the figure that, as the heat source temperature and evaporator pressure increases, the thermal efficiency of ORC increases. However, the effect of the heat source temperature is more noticeable in the case of ammonia compared to that of R134a. Figure 6 also shows that the maximum efficiency obtained was 14 % for R134a and 13% for ammonia at heat source temperature of 463K and evaporator pressure of 30 bars.

Figure 7 compares the Kalina cycle to the ORC using ammonia and R134a in terms of the thermal efficiency at heat source temperature of 373 K. Two ammonia / water concentration values were used; 0.66 and 0.55. It can be seen that the thermal efficiency of Kalina cycle with ammonia / water concentration of 0.55 is significantly higher than that of the ORC using ammonia and R134a at evaporator pressures below 20bars. For example at a pressure of 15bars, the KCS11 thermal efficiency (11.38%) with ammonia/water concentration of 0.55 is approximately 40% higher than that of the ORC using pure ammonia (7 %) and 20% higher than those of the ORC using pure R134a (9.2 %) with heat source temperature of 373K and heat sink temperature of 283K. At evaporator pressure higher than 20 bars, the

KCS11 thermal efficiency decreases significantly to be lower than those of the ORC. For the KCS11 with ammonia / water concentration of 0.66, its thermal efficiency is consistently higher than those of the ORC using ammonia over wide range of evaporator pressure values used but with similar values to those of the ORC using R134a. The high efficiency of the KCS11 at low operating pressure results in economic advantages in terms of lower turbine cost.

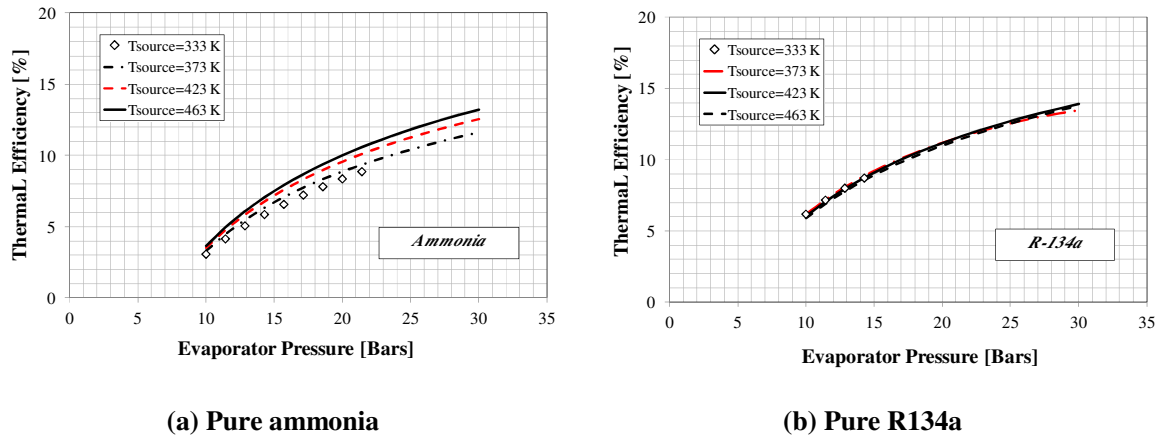


Figure 6: ORC thermal efficiency at various evaporator pressures and temperatures at heat sink temperature of 283K.

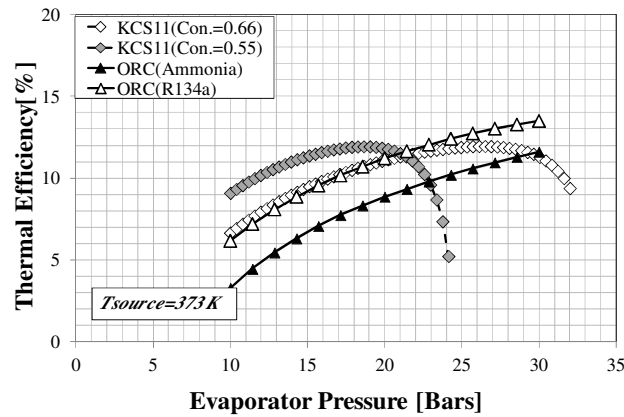


Figure 7: Comparison between Kalina cycle and ORC based on pure ammonia and pure R134a at evaporator temperature of 373 K and heat sink temperature of 283 K.

4. CONCLUSIONS

The performance of KCS11 using the ammonia-water mixture as working fluid was modelled and compared with the performance of ORC using pure ammonia or pure R134a as the working fluids. The results reveal that KCS11 with ammonia/water concentration of 0.55 achieves 20% to 40% higher efficiency than ORC, when using the same operating conditions of evaporator pressure of 15 bars, heat source temperature of 373K, and a heat sink temperature of 283K. The high efficiency of the Kalina cycle at low evaporator pressures will result in reducing the cost of the cycle components thus offsetting the cost of increasing the number of components and can lead to a cost effective power generation system. The results also show that at a given evaporator pressure, heat source and sink temperatures, an optimum ammonia mass fraction that can give maximum cycle efficiency can be determined.

NOMENCLATURE

h = Specific enthalpy (kJ/kg)

$\dot{m}$ = Mass flow rate (kg/s)

P = Pressure (bars)

q = Specific heat transfer (kJ/kg)

v = Specific volume (m^3/kg)

w = Specific work done (kJ/kg)

η = Efficiency (%)

ω = Mass fraction, $\dot{m}_7/\dot{m}_5$ (kg/kg)

SUBSCRIPTS

c = Condenser

e = Evaporator

net = Net output

pump = Pump

regen = Regenerator

turb = Turbine

s = Isentropic

1-10 = Cycle states, refer to Figure 1

ABBREVIATIONS

ORC = Organic Rankine Cycle

KCS11 = Kalina Cycle System11

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